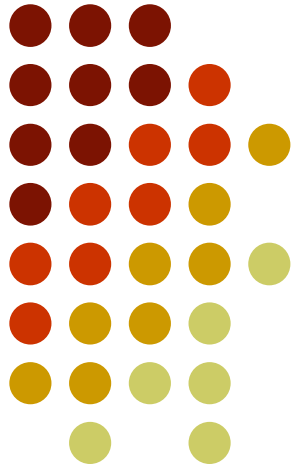
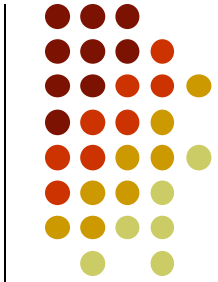
8 Arrow-Debreu经济及一般均衡



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概要



- Arrow-Debreu证券市场
- 状态价格
- 市场的完全性
- 参与者的优化
- 市场均衡

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8. 1Arrow-Debreu证券市场



- **状态ω或有要求权** state-ω contingent claim：状态为ω支付1，其他状态支付0的证券

- Arrow-Debreu证券：state-contingent security

- Arrow-Debreu证券市场：由所有的A-D证券，即他们的完全集合所构成的证券市场

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8. 1Arrow-Debreu证券市场

- 在A-D市场上，证券的个数等于可能的状态数，即N=Ω。对应的支付矩阵，即市场结构，是一个单位矩阵

$$X^{A-D}=\begin{pmatrix}1 & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & 1\end{pmatrix}=I$$

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8. 2状态价格

- ω状态索取权在0期的价格（以0期的消费品为单位）ϕ_ω
- 状态价格的完全集合已知
- 状态价格向量 $\phi \equiv [\phi_1; \cdots; \phi_\omega; \cdots \phi_\Omega]$

- 状态价格严格为正，否则意味着套利机会

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8. 3市场的完全性

- **完全市场**（complete market）：如果市场中的任一有限消费计划都可以通过有限成本的可交易证券的组合来融资，那么我们就称这个证券市场是完全的

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8. 3市场的完全性

- A-D证券市场是完全的
  - 任意消费计划

$$c = [c_1; \cdots c_\omega; \cdots; c_\Omega]$$

  - 状态ω或有证券组合

$$\theta = [c_1; \cdots c_\omega; \cdots; c_\Omega]$$

  - 终期支付

$$X^{A-D}\theta = [c_1; \cdots c_\omega; \cdots; c_\Omega] = c$$

  - 组合成本

$$p = \phi^T \theta$$

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8.3市场的完全性

- 例子
  - 假定未来有2个状态，市场上只有一只证券，未来支付为 $\begin{bmatrix}1\\1\end{bmatrix}$，是完全市场吗？
  - （接上例）假定市场上还有另一只证券，未来支付为 $\begin{bmatrix}2\\1\end{bmatrix}$，是完全市场吗？
  - 假定未来有2个状态，市场上有两只证券，未来支付为 $\begin{bmatrix}2\\1\end{bmatrix}$ 和 $\begin{bmatrix}4\\2\end{bmatrix}$，是完全市场吗？

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8. 4参与者的优化

- 参与者的禀赋
  - 假设1期禀赋e_1即0期证券的初始持有量
    - 复制组合：复制一个给定支付的组合
  - 金融财富：禀赋的总市值 $w = e_0 + e_1^T \phi$

- 预算约束：消费总成本=总财富

$$c_0 + \phi^T c_1 = w = e_0 + \phi^T e_1$$

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8. 4参与者的优化

$$\max_{c} \; u_{k,0}\left(c_{k,0}\right)+\rho_k \sum_{\omega=1}^{\Omega} p_\omega u_{k,1}\left(c_{k,1\omega}\right)$$

$$s.t. \quad c_{k,0}+\sum_{\omega=1}^{\Omega}\phi_\omega\left(c_{k,1\omega}-e_{k,1\omega}\right)=e_{k,0}$$

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8. 4参与者的优化

- 构建拉格朗日函数

$$L = u_{k,0}\left(c_{k,0}\right) + \rho_k \sum_{\omega=1}^{\Omega} p_\omega u_{k,1}\left(c_{k,1\omega}\right) + \lambda_k \left[e_{k,0} - c_{k,0} - \sum_{\omega=1}^{\Omega} \phi_\omega \left(c_{k,1\omega} - e_{k,1\omega} \right) \right]$$

- 一阶条件

$$\frac{\partial L}{\partial c_0} = 0: \quad u'_{k,0}\left(c_{k,0}\right) = \lambda_k$$

$$\frac{\partial L}{\partial c_{1\omega}} = 0: \quad \rho_k p_\omega u'_{k,1}\left(c_{k,1\omega}\right) = \lambda_k \phi_\omega, \forall \omega \in \Omega$$

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8. 4参与者的优化

- 消费

$$c_{k,0} = u'^{-1}_{k,0}(\lambda_k)$$

$$c_{k,1\omega} = u'^{-1}_{k,1}\left(\frac{\lambda_k \phi_\omega}{\rho_k p_\omega}\right)$$

- 代入预算约束

$$u'^{-1}_{k,0}(\lambda_k) + \sum_{\omega=1}^{\Omega} \phi_\omega \left[u'^{-1}_{k,1}\left(\frac{\lambda_k \phi_\omega}{\rho_k p_\omega}\right) - e_{k,1\omega} \right] = e_{k,0}$$

  - 可从中解出 λ_k

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8. 4参与者的优化

$$\frac{\rho_k p_\omega u'_{k,1}\left(c_{k,1\omega}\right)}{u'_{k,0}\left(c_{k,0}\right)} = \phi_\omega$$

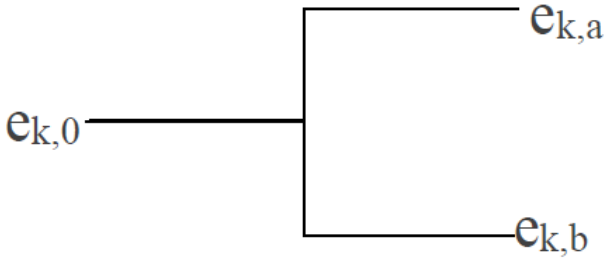
$$\frac{p_\omega u'_{k,1}\left(c_{k,1\omega}\right)}{p_{\omega'} u'_{k,1}\left(c_{k,1\omega'}\right)} = \frac{\phi_\omega}{\phi_{\omega'}}$$

- 状态价格之比等于不同时期/状态下消费的边际效用之比

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8. 4参与者的优化

- 算例：考虑一个经济，在1期有等概率的2个状态a和b，参与者k的禀赋如下



- 市场上有2个状态或有证券，价格分别为 ϕ_a 和 ϕ_b
- 参与者k具有如下形式的效用函数

$$U = u\left(c_{k,0}\right) + \frac{1}{2}\left[u\left(c_{k,a}\right) + u\left(c_{k,b}\right)\right]$$

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8. 4参与者的优化

$$\max_{c} \; u\left(c_{k,0}\right)+\frac{1}{2}\left[u\left(c_{k,a}\right)+u\left(c_{k,b}\right)\right]$$

$$s.t. \quad e_{k,0}+\phi_a e_{k,a}+\phi_b e_{k,b}-\left(c_{k,0}+\phi_a c_{k,a}+\phi_b c_{k,b}\right)=0$$

- 一阶条件

$$\frac{\partial L}{\partial c_{k,0}}=0: \quad u'\left(c_{k,0}\right)=\lambda_k$$

$$\frac{\partial L}{\partial c_{k,a}}=0: \quad \frac{1}{2}u'\left(c_{k,a}\right)=\lambda_k\phi_a$$

$$\frac{\partial L}{\partial c_{k,b}}=0: \quad \frac{1}{2}u'\left(c_{k,b}\right)=\lambda_k\phi_b$$

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8. 4参与者的优化

- 参与者1，效用函数形式为 $u_1 = \log(c)$

$$\frac{1}{c_0} = \lambda_1 \qquad c_{1,0} = \frac{1}{\lambda_1} \qquad c_{1,0} = \frac{w_1}{2}$$

$$\frac{1}{2} \times \frac{1}{c_{1,a}} = \lambda_1 \phi_a \qquad c_{1,a} = \frac{1}{2\lambda_1 \phi_a} \qquad c_{1a} = \frac{w_1}{2} \times \frac{1}{2\phi_a} = \frac{w_1}{4\phi_a}$$

$$\frac{1}{2} \times \frac{1}{c_{1,b}} = \lambda_1 \phi_b \qquad c_{1,b} = \frac{1}{2\lambda_1 \phi_b} \qquad c_{1b} = \frac{w_1}{2} \times \frac{1}{2\phi_b} = \frac{w_1}{4\phi_b}$$

- 代入预算约束，解得 $\lambda_1 = \frac{2}{w_1}$

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8. 4参与者的优化

- 参与者2，效用函数形式为 $u_2 = 2\sqrt{c}$

$$\frac{1}{\sqrt{c_{2,0}}} = \lambda_2 \qquad c_{2,0} = \frac{1}{\lambda_2^2} \qquad c_{2,0} = \frac{w_2}{1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}}$$

$$\frac{1}{2\sqrt{c_{2,a}}} = \lambda_2 \phi_a \qquad c_{2,a} = \frac{1}{4\lambda_2^2 \phi_a^2} \qquad c_{2,a} = \frac{w_2}{4\phi_a^2\left(1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}\right)}$$

$$\frac{1}{2\sqrt{c_{2,b}}} = \lambda_2 \phi_b \qquad c_{2,b} = \frac{1}{4\lambda_2^2 \phi_b^2} \qquad c_{2,b} = \frac{w_2}{4\phi_b^2\left(1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}\right)}$$

- 代入预算约束，解得 $\frac{1}{\lambda_2^2} = \frac{w_2}{1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}}$

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8.5市场均衡

- Arrow-Debreu经济的均衡是使如下条件成立的状态价格集合

1. 给定状态价格，每一参与者达到最优化

$$c_k\left(e_k,\phi\right)=Arg\max U_k\left(c_k\right)$$

$$s.t.\ \ c_{k,0}+\phi^T c_{k,1}=e_{k,0}+\phi^T e_{k,1}$$

2. 证券市场出清

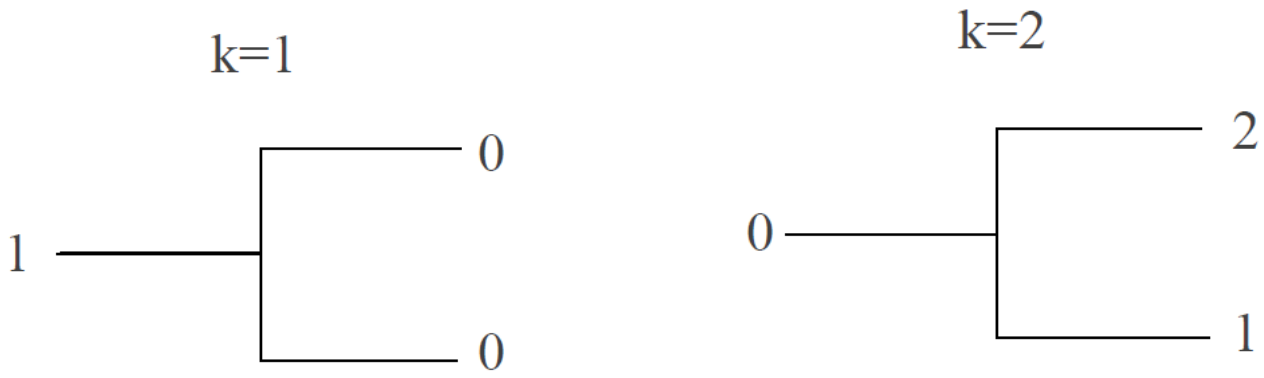
$$\sum_k c_{k,0}\left(e_k,\phi\right)=\sum_k e_{k,0}$$

$$\sum_k c_{k,1}\left(e_k,\phi\right)=\sum_k e_{k,1}$$

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8.5市场均衡

- 沿用前例
- 参与者1和2的禀赋如下



- 则其禀赋的市值为 $w_1 = 1, w_2 = 2\phi_a + \phi_b$

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8.5市场均衡

- 市场出清

$$1 = c_{1,0} + c_{2,0} = \frac{1}{2} + \frac{2\phi_a + \phi_b}{1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}} = \frac{1}{2} + \frac{2\phi_a + \phi_b}{\alpha}$$

$$2 = c_{1,a} + c_{2,a} = \frac{1}{4\phi_a} + \frac{2\phi_a + \phi_b}{4\phi_a^2\left(1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}\right)} = \frac{1}{4\phi_a} + \frac{2\phi_a + \phi_b}{4\alpha\phi_a^2}$$

$$1 = c_{1,b} + c_{2,b} = \frac{1}{4\phi_b} + \frac{2\phi_a + \phi_b}{4\phi_b^2\left(1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}\right)} = \frac{1}{4\phi_b} + \frac{2\phi_a + \phi_b}{4\alpha\phi_b^2}$$

记 $\alpha \equiv 1 + \frac{1}{4\phi_a} + \frac{1}{4\phi_b}$

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8. 5市场均衡

$$\frac{2\phi_a + \phi_b}{\alpha} = \frac{1}{2}$$

$$\frac{1}{4\phi_a} + \frac{1}{8\phi_a^2} = 2$$

$$\phi_a = \frac{1+\sqrt{17}}{16} \approx 0.32$$

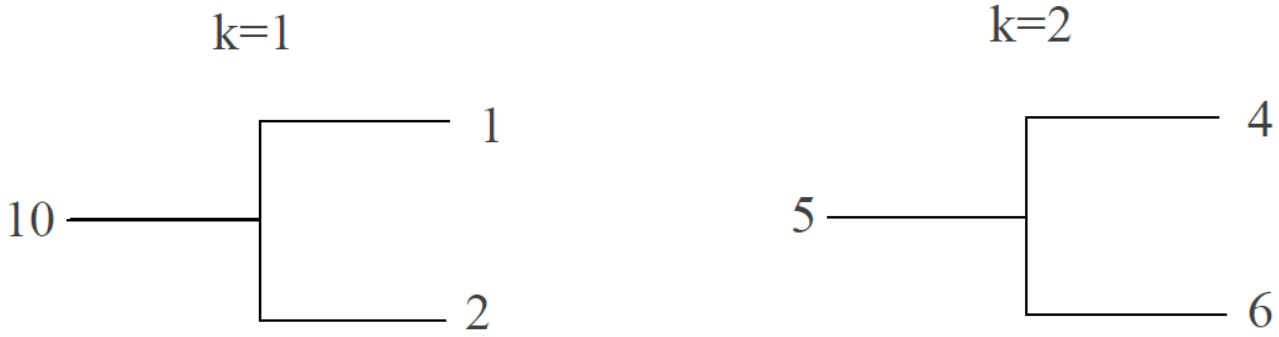
$$\frac{1}{4\phi_b} + \frac{1}{8\phi_b^2} = 1$$

$$\phi_b = \frac{1}{2}$$

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均衡算例

- 考虑一个经济，在1期有两个状态a和b
- 经济中有两个参与者，其禀赋为



- 参与者1和2的偏好都可以用如下效用函数表示

$$U_k = \frac{1}{2} c_{k,0} + 0.9\left(\frac{1}{3}\ln c_{k,a} + \frac{2}{3}\ln c_{k,b}\right)$$

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均衡算例

- 参与者的优化
  - 以参与者1为例

$$\max U_1 = \frac{1}{2}c_{1,0} + 0.9\left(\frac{1}{3}\ln c_{1,a} + \frac{2}{3}\ln c_{1,b}\right)$$

$$s.t.\ \ c_{1,0} + \phi_a c_{1,a} + \phi_b c_{1,b} = 10 + \phi_a + 2\phi_b$$

  - 构建拉格朗日函数

$$L = \frac{1}{2}c_{1,0} + 0.9\left(\frac{1}{3}\ln c_{1,a} + \frac{2}{3}\ln c_{1,b}\right) + \lambda_1\left(10 + \phi + 2\phi_b - c_{1,0} - \phi_a c_{1,a} - \phi_b c_{1,b}\right)$$

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均衡算例

- 一阶条件

$$\frac{\partial L}{\partial c_{1,0}} = 0: \frac{1}{2} - \lambda_1 = 0$$

$$\frac{\partial L}{\partial c_{1,a}} = 0: \ 0.9 \times \frac{1}{3} \times \frac{1}{c_{1,a}} - \lambda_1 \phi_a = 0$$

$$\frac{\partial L}{\partial c_{1,b}} = 0: \ 0.9 \times \frac{2}{3} \times \frac{1}{c_{1,b}} - \lambda_1 \phi_b = 0$$

- 解得 $\lambda_1 = \frac{1}{2}, c_{1,a} = \frac{0.6}{\phi_a}, c_{1,b} = \frac{1.2}{\phi_b}$

- 参与者2的优化分析类似，解得 $\lambda_2 = \frac{1}{2}, c_{2,a} = \frac{0.6}{\phi_a}, c_{2,b} = \frac{1.2}{\phi_b}$

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均衡算例

- 市场出清

$$5 = c_{1,a} + c_{2,a} = \frac{1.2}{\phi_a} \qquad \phi_a = 0.24$$

$$8 = c_{1,b} + c_{2,b} = \frac{2.4}{\phi_b} \qquad \phi_b = 0.3$$

- 均衡消费
  - 参与者1：$c_{1,0} = 9.04, c_{1,a} = 2.5, c_{1,b} = 4$
  - 参与者2：$c_{2,0} = 5.96, c_{2,a} = 2.5, c_{2,b} = 4$

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8.6一般均衡与局部均衡

- 考虑一个经济，未来有等概率的2个状态，市场中有两个证券A和B，未来支付如下



- 假定无风险利率为0，A和B的价格应为多少？
- CAPM：0.5

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8.6一般均衡与局部均衡

- 假设经济中只有一位参与者，偏好如下

$$u = \ln c_0 + \frac{1}{2}\left(\ln c_a + \ln c_b\right)$$

- 禀赋：0期的1单位消费品，α单位证券A和β单位证券B

$$1 + \alpha\phi_a + \beta\phi_b$$

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8. 6一般均衡与局部均衡

$$\max_{c} \ \ln(c_0)+\frac{1}{2}\left[\ln(c_a)+\ln(c_b)\right]$$

$$s.t. \quad 1+\phi_a\alpha+\phi_b\beta-(c_0+\phi_a c_a+\phi_b c_b)=0$$

$$L=\ln(c_0)+\frac{1}{2}\left[\ln(c_a)+\ln(c_b)\right]+\lambda\left[1+\phi_a\alpha+\phi_b\beta-(c_0+\phi_a c_a+\phi_b c_b)\right]$$

$$\frac{\partial L}{\partial c_0}=0: \frac{1}{c_0}=\lambda \qquad c_0=\frac{1}{\lambda}=1 \qquad \lambda=1$$

$$\frac{\partial L}{\partial c_a}=0: \ \frac{1}{2c_a}=\lambda\phi_a \qquad c_a=\frac{1}{2\lambda\phi_a}=\alpha \qquad \phi_a=\frac{1}{2\alpha}$$

$$\frac{\partial L}{\partial c_b}=0: \ \frac{1}{2c_b}=\lambda\phi_b \qquad c_b=\frac{1}{2\lambda\phi_b}=\beta \qquad \phi_b=\frac{1}{2\beta}$$

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8.6一般均衡与局部均衡

- 无风险证券价格

$$\phi_a + \phi_b = \frac{1}{2\alpha} + \frac{1}{2\beta} = \frac{\alpha + \beta}{2\alpha\beta}$$

- 无风险利率为0，有

$$\alpha + \beta = 2\alpha\beta$$

- A、B的价格不一定总是0.5！

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